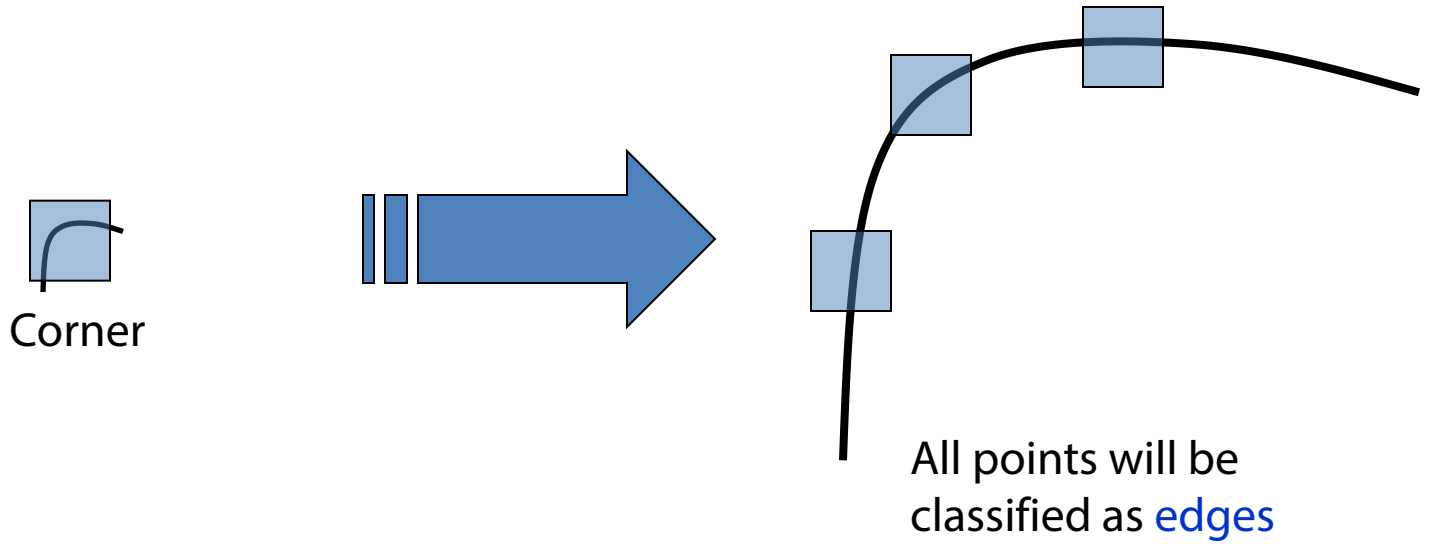
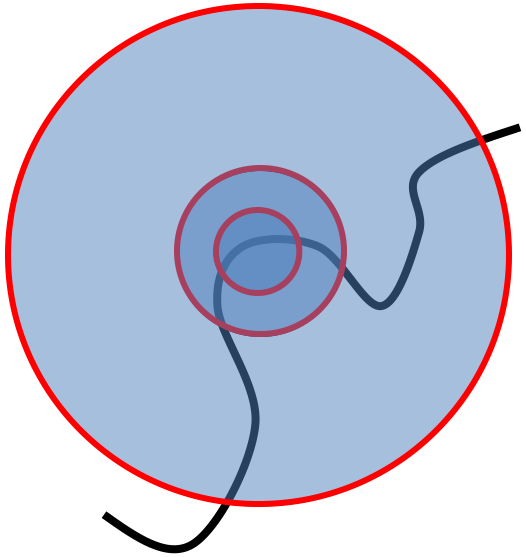
Harris detector invariance properties: scaling



Neither invariant nor equivariant to scaling

Scale invariant detection

Suppose you're looking for corners



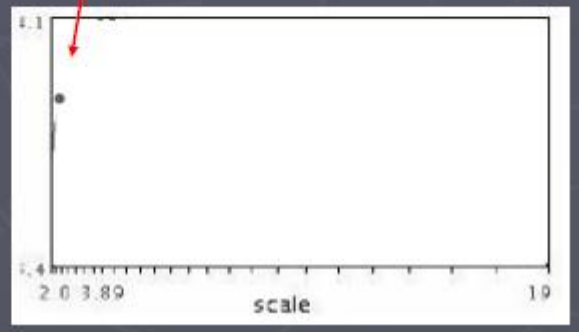
Key idea: find scale that gives local maximum of *f*

- in both position and scale
- One definition of *f*: the Harris operator

Automatic scale selection

Lindeberg et al., 1996



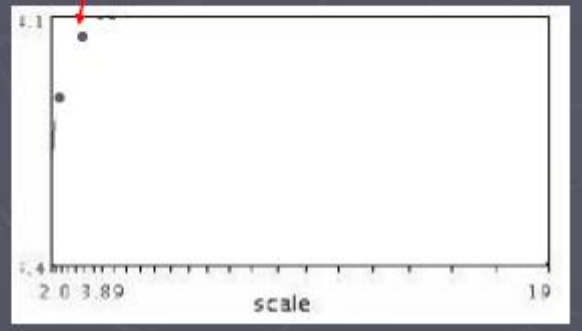


$$f(I_{i_1 \ldots i_m}(x,\sigma))$$

Slide from Tinne Tuytelaars

Automatic scale selection

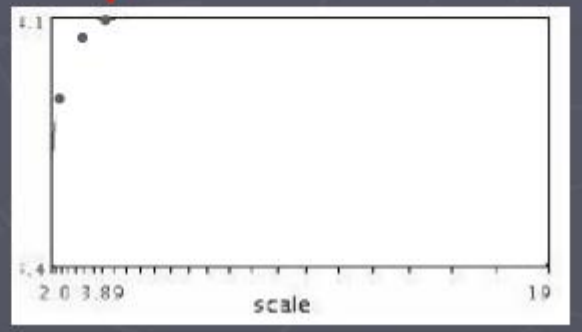




$$f(I_{i_1 \ldots i_m}(x,\sigma))$$

Automatic scale selection

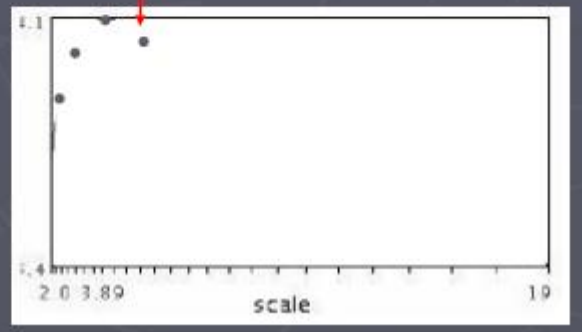




$$f(I_{i_1 \ldots i_m}(x,\sigma))$$

Automatic scale selection

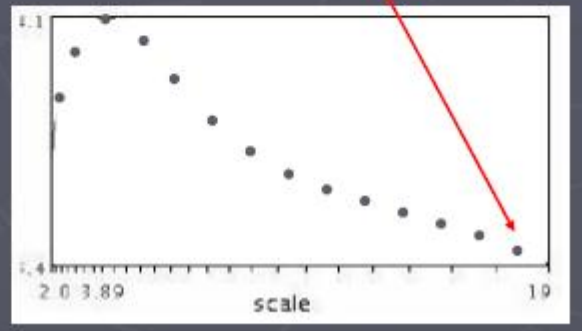




$$f(I_{i_1 \ldots i_m}(x,\sigma))$$

Automatic scale selection

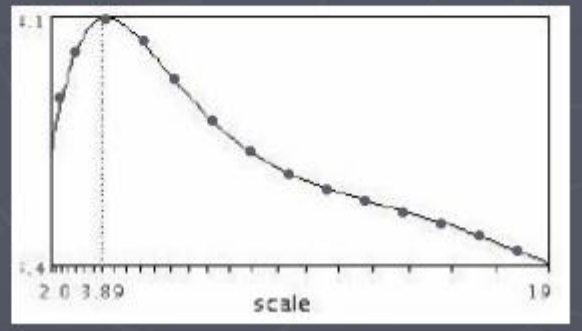




$$f(I_{i_1 \ldots i_m}(x,\sigma))$$

Automatic scale selection





$$f(I_{i_1 \ldots i_m}(x,\sigma))$$